

Online Social Network Analysis

Homework 4

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Roseland - CA 49

1. Data-based characterization of Roseland

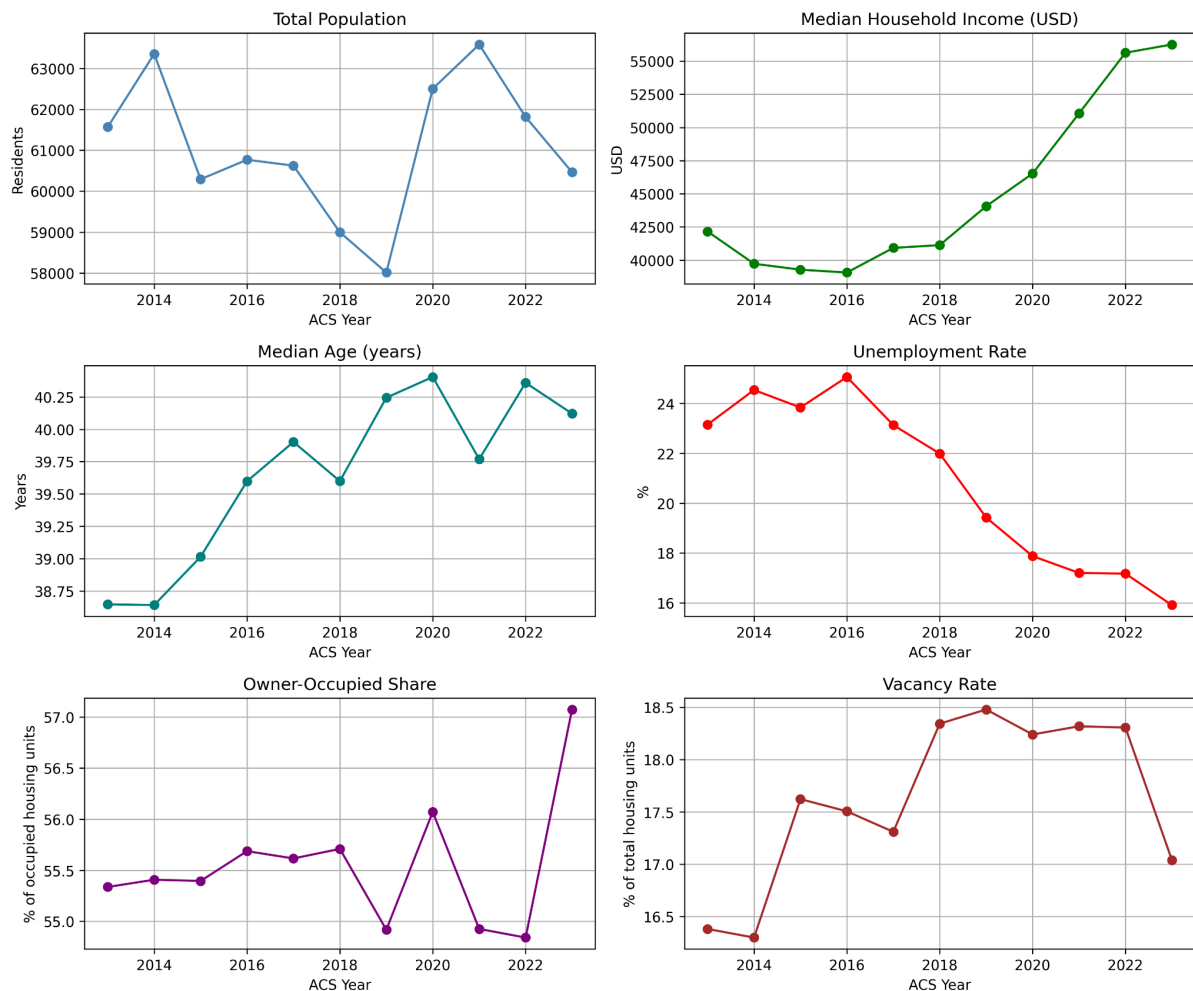
Overview

Roseland is a predominantly black community on the Far South Side of Chicago. Using ACS block group data and decennial census block data, we profile demographic, housing, and labor market trends from 2013 to 2023, and provide 2010 vs 2020 decennial comparisons for level checks.

Data caution

Estimates come from ACS 5-Year Data at the block group level. Due to small sample sizes, year to year variation does not necessarily reflect true change. I smooth key series with a centered 3-year rolling mean and note large inter-annual swings where relevant.

Data analysis



Population and age: Roseland's overall population from 2013 to 2023 ranged from approximately 58,000 to 63,500 residents. Minor multi-year oscillations likely represent sampling variability within ACS block group estimates but the decline around 2019–2020 coincides with the COVID-19 pandemic, which disrupted household mobility and data gathering. The slight downward trend could

also be a signal of longer-term out-migration from the Far South Side, where population decline has persisted since the 1990s due to job loss, disinvestment, and declining birth rates.

The median age increased from about 38.7 years to just above 40 during 2020–2022, leveling off afterward. The rise represents gradual aging of the resident population, consistent with statewide trends of aging baby boomers and declining younger in-migrants. In Roseland's case, this would presumably reflect aging in place by older homeowners and out-migration of younger adults and families to seek employment opportunities or newer houses elsewhere in the metro area.

Income and labor market: Median household income declined somewhat through 2016 to about \$39,000, then rose steadily to nearly \$56,000 by 2023. This trend mirrors broader post-recession restoration and tightening labor markets in Chicago pre- and post-pandemic. The trend suggests some recovery in area compensation levels, potentially augmented by citywide minimum wage increases and service-sector recovery.

Concurrently, unemployment fell sharply from roughly 24–25% in the mid-2010s to nearly 16% in 2023, with the largest decline being between 2018 and 2020. This likely reflects economic stabilization, infrastructure investment, and social programs targeting South Side communities, though Roseland's unemployment rate continues to far exceed the city average (around 6–7%), indicating long-standing labor market disadvantage.

Labor force participation increased from 44% to about 47% during 2018–2019 before falling somewhat by 2023. This development could reflect short-term advances in labor force participation that fizzled out as pandemic stress and structural issues (e.g., availability of transportation and skill imbalance) reemerged.

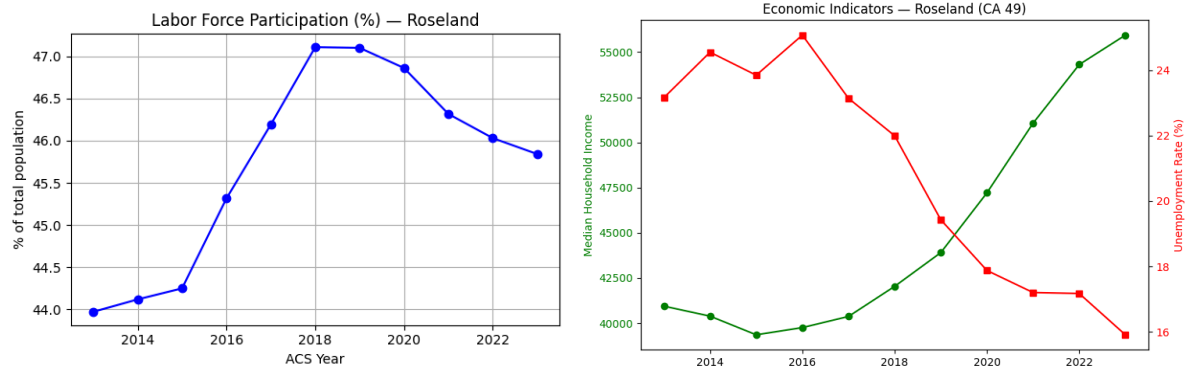
Together, these labor and income indicators indicate consistent economic advancement, yet with ongoing struggle to hold on to jobs and fair income growth relative to the rest of Chicago.

Housing: The homeownership by share owner remained in the mid-50s, rising a bit in 2023. This stability indicates a nucleus of long-term owners who are maintaining tenure amidst other market unpredictability. Roseland's high homeownership relative to other South Side communities is grounded in its heritage as a working-class residential community, although more current affordability issues may limit new buyers.

The vacancy rate of 16–18% remains higher than the citywide rate in Chicago (around 10%). The high vacancy is due to ongoing housing disinvestment and population loss, as well as the lingering effect of the 2008 foreclosure crisis that most impacted South Side communities. The modest increase between 2018–2021 could be pandemic-era aberration and delayed recovery of properties, with stabilization as economic conditions recovered by 2023.

Overall, housing data reflect local demand structural weakness, notwithstanding strengthened income and employment. Chronic vacancy and lack of ownership gains reflect weak turnaround and disuniformed recovery in Roseland's housing market.

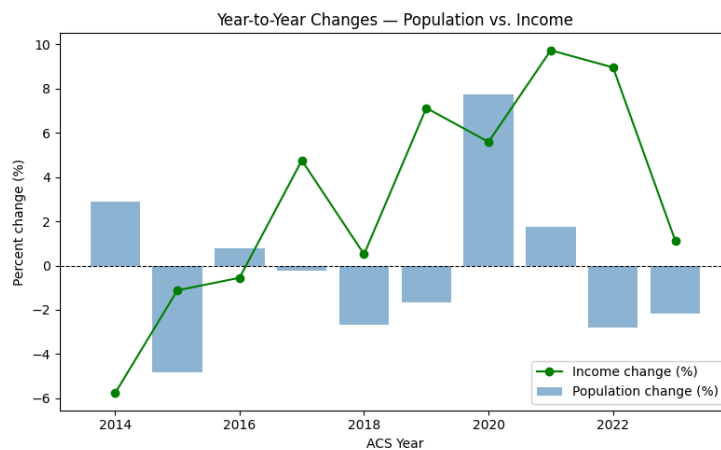
Economic indicators



These figures demonstrate the clear inverse correlation between median household income and the unemployment rate from 2013 to 2023. Rates of income plateaued in the early 2010s and began a strong rebound after 2016, increasing from approximately \$39,000 to \$56,000, as unemployment fell from around 25% to 16%. The break-even point approximately at 2018–2020 is a significant shift in Roseland's local economy, likely the result of aggregate post-recession restoration and local reinvestment efforts.

The parallel trends have shown that employment gains immediately transferred to higher household income, an increasingly balanced but still uneven neighborhood labor market. In spite of the parallel trends, the unemployment rate of Roseland remains substantially above the Chicago standard, reflecting continuing structural barriers such as limited access to employment, educational disadvantages, and transportation handicaps.

Temporal change analysis

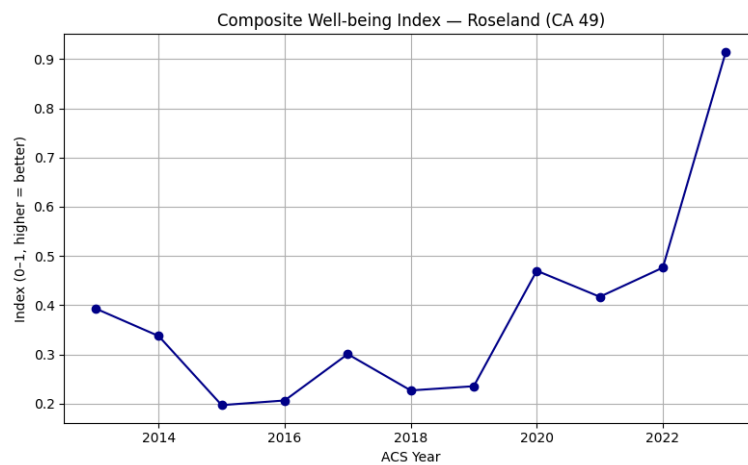


This chart compares annual percentage change in population (bars) and median income (line) of Roseland. Population change fluctuates around zero, with modest drops in some years and modest rises around 2018–2020, consistent with a relatively stable but not growing population base. Income change is more volatile in its growth, hitting nearly 10% yearly growth around 2021, consistent with very rapid economic catch-up following prior stagnation.

The lack of correspondence between population stability and income growth implies that the gains are being realized among existing residents rather than through in-migration, and that suggests rising prosperity for those who remain but little replenishment of the population. This process is equivalent

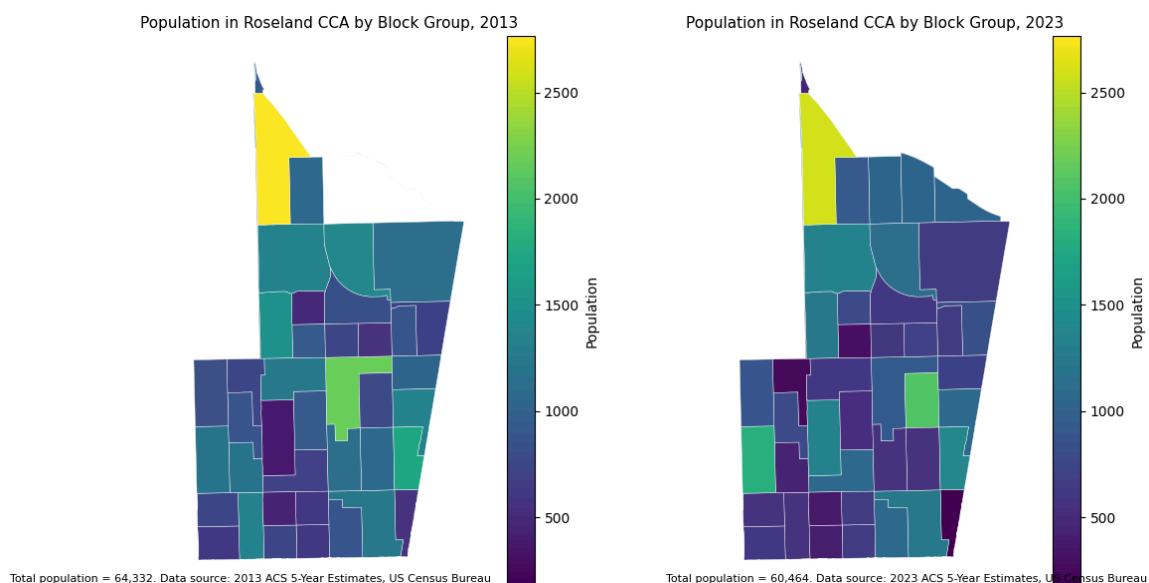
to selective out-migration of lower-income households and holding onto older, more established homeowners.

Well-being index



The composite index integrates income, unemployment, and homeownership into a single 0–1 score with higher values indicating greater socioeconomic well-being. Roseland's index fluctuates in the lower range (0.2–0.4) between 2013 and 2018, reflecting a decade of relative economic stagnation. Since 2018, the index spikes violently, to over 0.9 by 2023, mirroring the strong household income bounce back and downward trending unemployment rates. This violent spike suggests extreme post-2018 socioeconomic progress, likely due to overall economic growth, wage improvements, and fiscal support during the pandemic period. Despite the strong upward momentum, Roseland's root issues of an aging population base, high vacancy rates, and limited new investment mean that maintaining such progress will require ongoing public and private reinvestment.

Decennial comparison



Between 2013 and 2023, Roseland's total population declined from approximately 64,000 to 60,000 residents, a loss of about 6% over the decade. The spatial pattern shows that population decreases were widespread, with the sharpest declines in several central and southern block groups, while a few northern and western areas remained relatively stable. This modest but persistent decline aligns with

long-term depopulation trends across many South Side neighborhoods, reflecting continued out-migration, aging households, and limited new housing growth.

Key takeaways

- Population: Stable but slowly declining ($\approx 58k-63k$), with an aging population and continued out-migration.
- Demographics: Still over 95% Black, showing minor diversification by 2020.
- Economy: Household income rose from \$39k to \$56k as unemployment fell from 25% to 16%, reflecting strong post-2018 recovery but still above citywide joblessness.
- Housing: Homeownership steady near 55%, but vacancy high (16–18%), indicating weak demand and slow reinvestment.
- Well-being: The composite index surged after 2018, showing economic rebound yet with persistent structural barriers and aging infrastructure.

Overall: Roseland is economically improving but demographically stagnant, needing sustained investment to translate recovery into lasting neighborhood vitality.

2. Descripton of the data (i.e. variables) chosen to use as the basis of the study along with a discussion of why you chose these variables and any preliminary analysis you did to narrow the data

Variables:

Name	Database	Description
Total population	B01003_001E	Base for rates and shares
Median House income	B19013_001E	Income and purchasing power proxy
Median Age	B01002_001E	Life cycle and aging structure
Unemployed rate	B23025_005E / B23025_003E	Local labor slack
Owner-occupied share	B25003_002E / B25003_002E	Household stability proxy
Vacancy rate	B25002_003E / B25002_001E	Housing demand or distress signal
Percent Black	B02001_003E / B02001_003E	Racial composition and change

Data cleaning

- Replace nonpositive or error codes with NaN for all count and estimate fields, then aggregate by year.

- For medians and ratios that vary with population size, compute population-weighted means across block groups using B01003_001E as weights.
- After aggregation, compute rates and shares, then apply a centered 3-year rolling mean to smooth sampling noise: total_pop, median_income, labor_force, unemployed.
- Guard against divide by zero by filtering years with labor_force greater than 0.
- Keep decennial counts as reported, no smoothing.

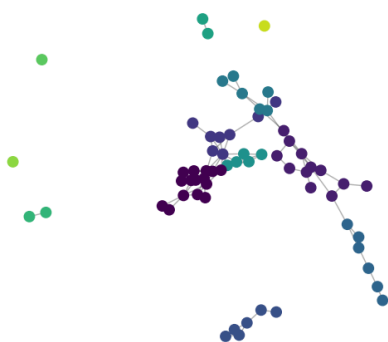
3. Block groups network

(a) Modeling and analysis

To analyze the internal cohesion of Roseland, the block groups comprising the Community Area, were modeled as socioeconomic similarity network. The socioeconomic profile of each block group was characterized using indicators from ACS, including median household income, owner-occupied housing rate, unemployment rate, and housing vacancy rate. These variables were selected because they jointly capture economic stability and housing labor conditions.

All indicators were normalized to have a standard range by using Min-Max scaling. For interpretability, all negative indicators such as unemployment and vacancy rates were inverted so that higher values reflect better socioeconomic conditions. Pairwise euclidean distances between block groups were then computed in this normalized feature space. To model similarity relationships, edges were added between each block group and its k most similar neighbors. (threshold set to 0.75)

Socioeconomic Similarity Graph — Roseland Block Groups



The resulting graph captures how block groups cluster naturally according to shared socioeconomic profiles, rather than being constrained by geographic adjacency.

- This analysis identified nine connected components in the network, each representing a set of block groups with internally strong similarity ties. The largest connected component corresponds to the dominant socioeconomic cluster in Roseland.

(b) Why Roseland Is a Community

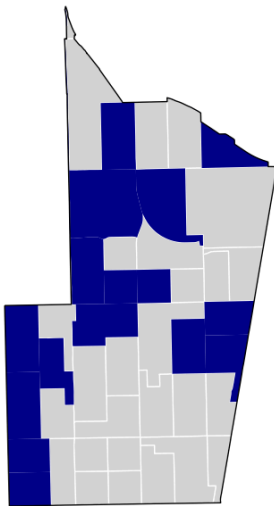
The socioeconomic similarity network reveals a large, dense connected component encompassing 28 block groups, which together form the cohesive core of Roseland. Within this cluster, block groups display short similarity distances and strong internal ties, indicating consistent socioeconomic conditions across income, housing ownership, and employment. On the map, this cluster appears as a geographically continuous zone spanning the northern and central portions of the community area. The relative uniformity of these indicators suggests shared economic realities and potentially overlapping social and institutional ties. Overall, the data support the view that Roseland possesses a socioeconomically cohesive core, functioning as a community characterized by internal similarity and interconnectedness.

(c) Why Roseland Is Not a Single Community

While one central cluster is quite cohesive in Roseland, the network is fragmented into nine overall connected components, many of which are small and only weakly connected to the main group. These peripheral block groups also show distinct socioeconomic patterns—either much more affluent or much more disadvantaged than the core—and appear to be either spatially dispersed or socially detached from the map. Several disconnected clusters and isolated nodes suggest the presence of socioeconomic segmentation in the CA; some areas must be quite different in terms of income, housing stability, or employment conditions. Therefore, although Roseland has a well-defined socioeconomic center, it cannot be considered to represent a unified community but rather a set of smaller, internally cohesive subcommunities with varying levels of attachment to the core.

(d) Proposal of new community

Largest Socioeconomic Similarity Cluster — Roseland



The socioeconomic similarity network revealed clear internal differentiation within Roseland, indicating that the current community area boundaries do not fully align with patterns of socioeconomic cohesion.

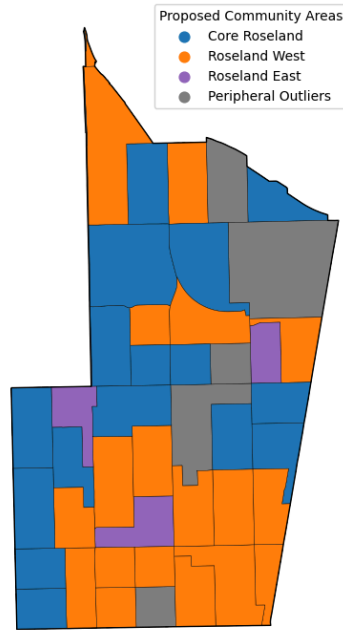
First, a two-group division based on the largest connected component highlights a core cluster of highly similar block groups surrounded by more diverse or disconnected areas. This division provides strong evidence that Roseland does not function as a single cohesive socioeconomic community, but rather contains a central region of internal similarity and a surrounding set of more heterogeneous neighborhoods.

Building on this insight, a finer-grained reorganization identifies four distinct subcommunities:

1. A Core Roseland area composed of internally cohesive block groups with similar economic and housing characteristics;
2. Two transitional zones (West and East) that bridge the core and the periphery, representing areas of moderate similarity to the core; and
3. Several peripheral outlier clusters that form small, socioeconomically distinct pockets within or adjacent to the main community.

This structure provides a data-driven alternative organization for Roseland, one that better captures the diversity of local socioeconomic conditions and internal cohesion patterns. It suggests that community planning and policy efforts should recognize Roseland not as a single, uniform area but as a collection of interconnected subcommunities with varying degrees of similarity and need.

Proposed Socioeconomic Subcommunities — Roseland



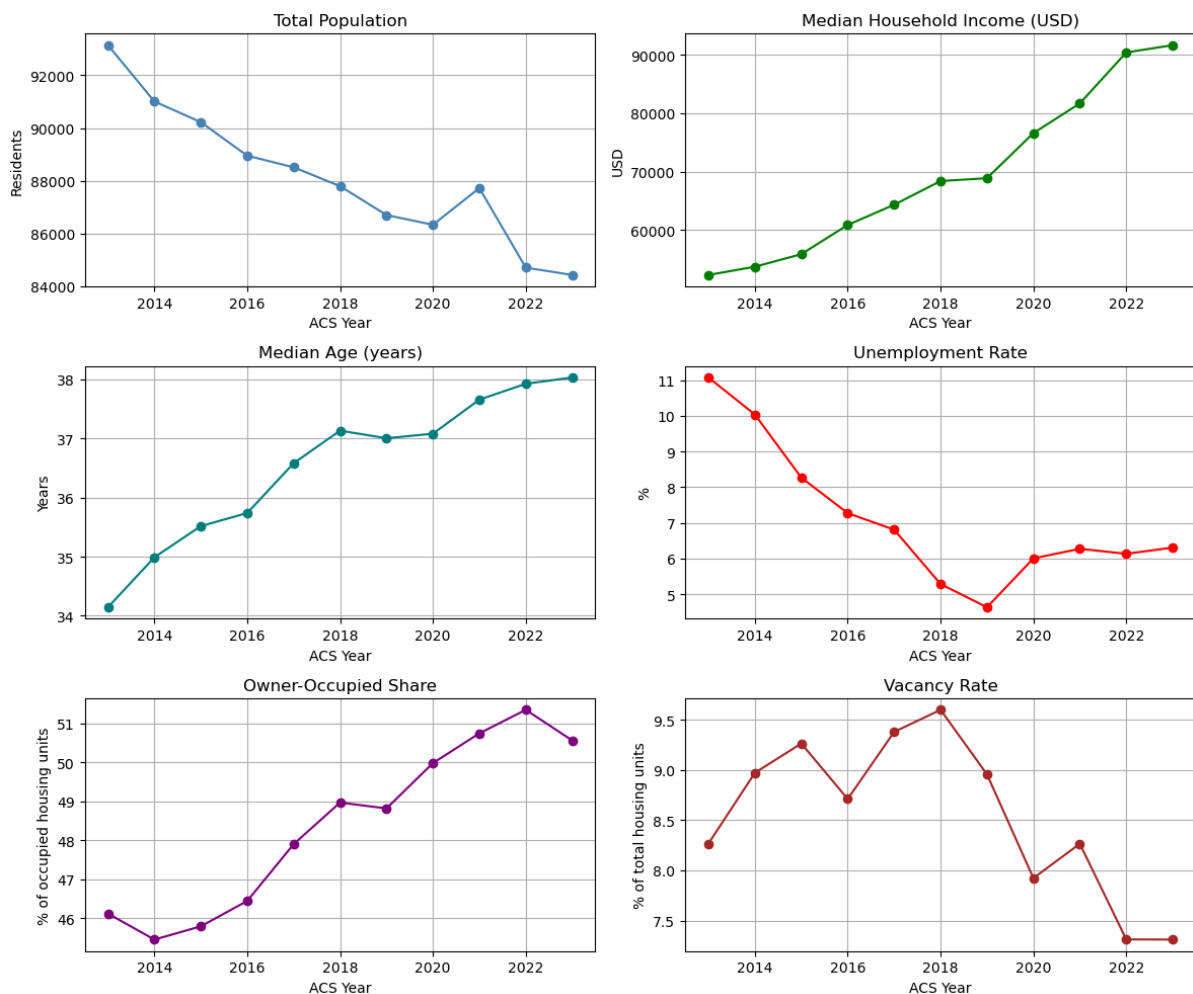
Irving Park - CA 16

1. Data-based characterization of Irvine Park

Overview

Irving Park is a historic community area on Chicago's Northwest Side, centered on the Villa District and Independence Park. It mixes early twentieth century homes, classic bungalows, two flats, and newer infill, with strong transit links and steady neighborhood character.

Data analysis



Population: Irving Park shows a steady population decline from roughly 93,000 residents in 2013 to about 84,500 in 2023, with a brief uptick around 2021 followed by a sharper dip in 2022. The overall slide suggests continued out-migration and smaller household sizes, while the 2022 drop likely reflects post-pandemic mobility shifts and ACS sampling noise. Fewer residents can soften local retail demand and school enrollment, even as it creates openings for reuse of properties or infill efforts.

Age: The median age climbs from about 34 years in 2013 to around 38 in 2023, with modest plateaus in the late 2010s. The pattern is consistent with aging in place among established homeowners combined with fewer young in-migrants and families. An older age profile implies growing demand

for health, accessibility, and aging-friendly housing, while strategies to attract younger households could help rebalance the demographic mix.

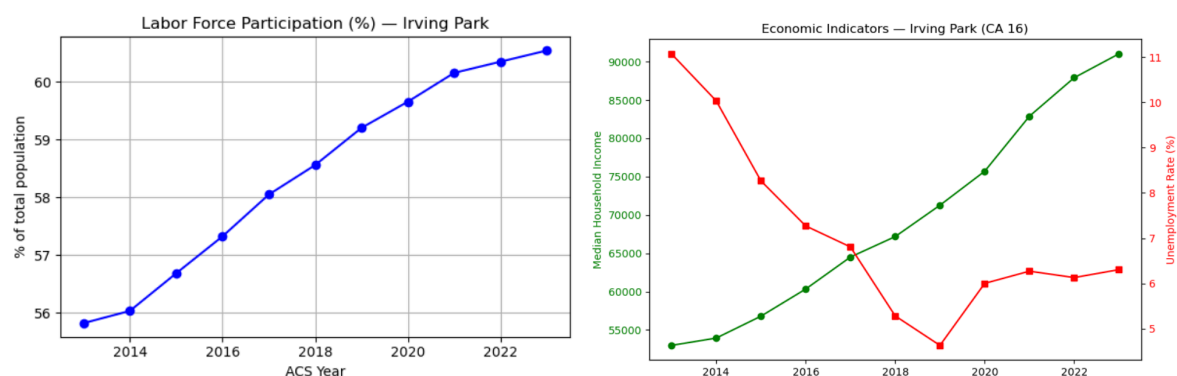
Household Income: Median household income rises almost continuously from approximately \$53,000 in 2013 to just over \$90,000 by 2023, with particularly strong gains after 2016 and again from 2020 to 2022. This trajectory points to improving earnings and possible compositional change toward higher-income households, aided by citywide wage growth and sectoral recovery. The upside for neighborhood commerce is clear, but the pace also raises affordability concerns for long-time renters and lower-income residents.

Unemployment Rate: Unemployment falls sharply from roughly 11% in 2013 to a low near 4.7% in 2019, then settles in the 6–6.3% range during 2020–2023. The pre-pandemic improvement suggests strong labor absorption, and the post-pandemic level indicates partial normalization without fully reclaiming the 2019 trough. Linking residents to jobs through transit access, skills training, and targeted placement programs remains central to regaining earlier lows.

Owner-Occupied Share: Owner occupancy trends upward from about 46% in 2013 to a peak near 51% in 2022, edging slightly lower in 2023 but remaining above its starting point. This signals a stable core of long-term owners and incremental strengthening of homeownership during the decade. Sustaining these gains will depend on accessible mortgage products, support for first-time buyers, and resources for home maintenance and rehabilitation.

Vacancy: Vacancy peaks around 9.6% in 2018 and then declines steadily to roughly 7.3% by 2022–2023. The improvement implies healthier absorption and a firmer rental market relative to mid-decade conditions, with post-pandemic stabilization evident in the most recent years. Continued progress will likely come from rehabilitating lingering vacant units and pairing redevelopment with affordability measures to ensure existing residents benefit from the tightening market.

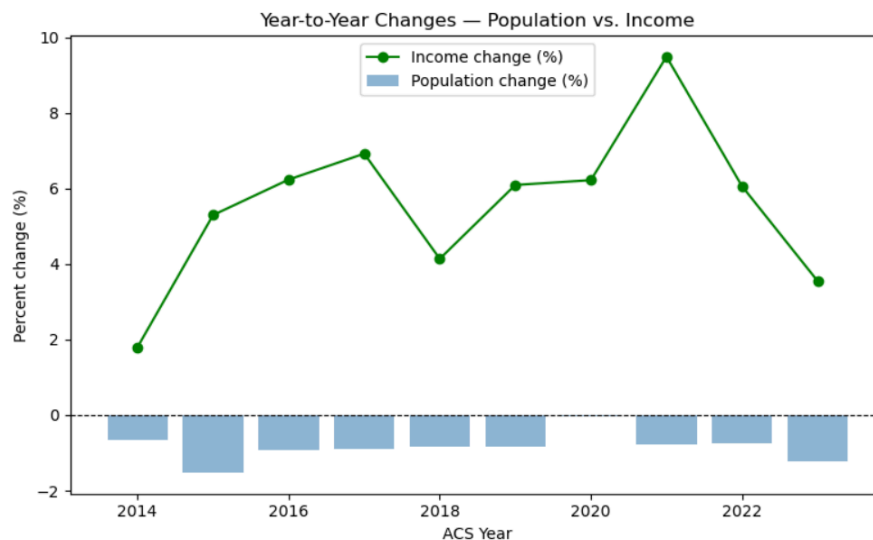
Economic indicators



Labor force participation trends upward throughout the period, increasing from about 55.8% in 2013 to roughly 60.5% in 2023, with steady year-over-year gains and only minor slowing after 2021. The sustained rise indicates more residents are either working or actively seeking work, which can reflect demographic shifts toward working-age adults, better attachment to the labor market, or expanded opportunities that draw people in. When paired with higher median incomes and a post-2019 unemployment rate that remains somewhat elevated, the data suggest Irving Park is engaging a larger share of its population in the labor market while still facing barriers to full employment, issues that targeted training, transportation links, and job placement could help address.

The chart shows a clear decoupling between income gains and unemployment over the decade. Median household income rises from roughly \$53,000 in 2013 to a little above \$91,000 by 2023, with the slope steepening after 2016 and again from 2020 to 2022. In contrast, unemployment drops from about 11% in 2013 to a trough near 4.7% in 2019, then rebounds during the pandemic and settles around 6–6.3% through 2020–2023. Taken together, the pattern suggests strong wage growth and recovery in higher-paying segments alongside an incomplete return to pre-pandemic labor tightness. For Irving Park, this combination points to improving household earnings and spending power, but also to persistent frictions in job matching and access that keep unemployment above the 2019 low.

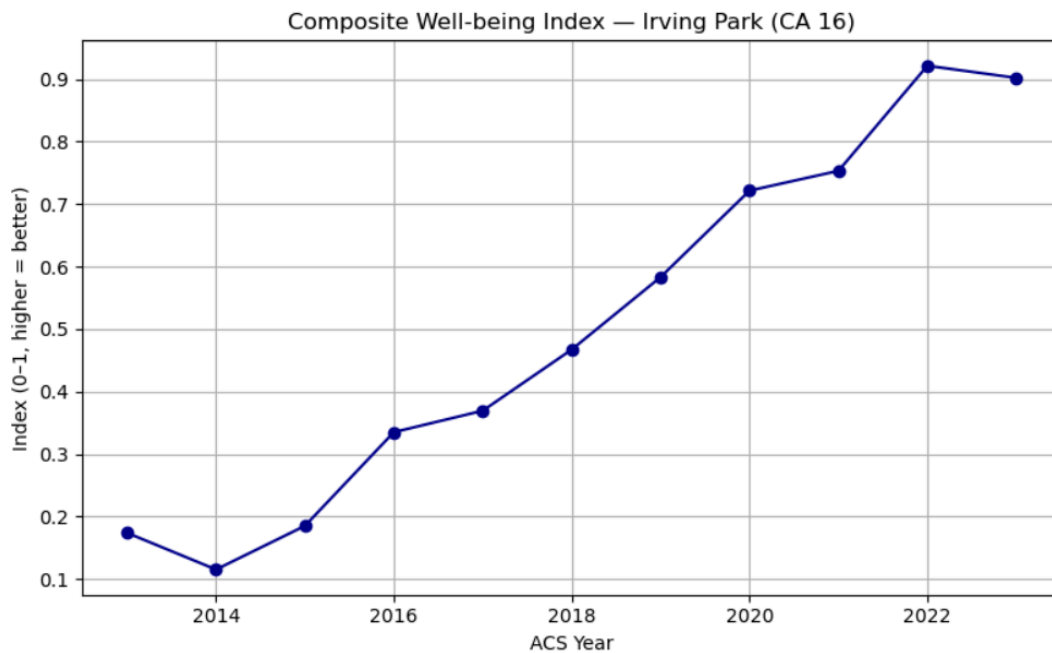
Temporal change analysis



The chart highlights a persistent divergence: incomes rise every year while population edges down slightly each year. Median household income posts positive annual gains throughout the period roughly five to seven percent in the mid-2010s, a softer increase around 2018, renewed growth near six percent in 2019–2020, a surge close to ten percent in 2021, and then moderating to about six percent in 2022 and roughly three to four percent in 2023. At the same time, the population bars remain below zero each year, indicating small but steady declines typically around half to a little over one percent annually.

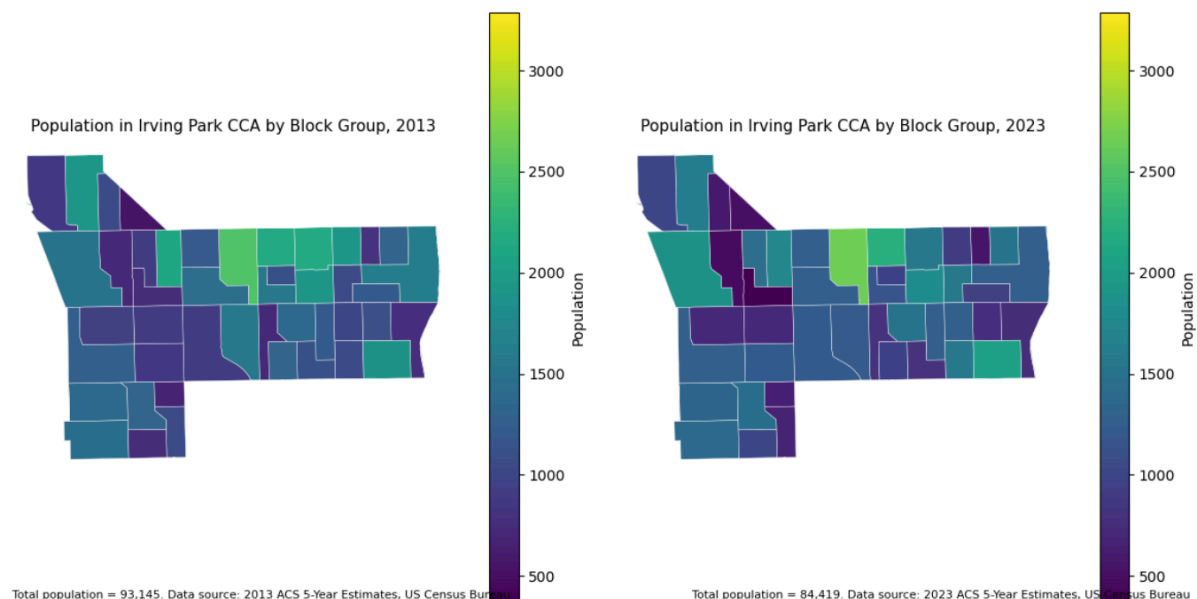
Together, these dynamics suggest that Irving Park is experiencing income growth faster than headcount change would predict, likely reflecting a mix of rising wages, compositional shifts toward higher-earning households, and continued out-migration among lower-income residents. The result is higher per-capita resources alongside potential affordability pressure and a changing customer base for local businesses, which could tighten the rental market even as total demand for some neighborhood services softens.

Well-being index



The composite well being index for Irving Park rises strongly over the decade. It starts near 0.17 in 2013, dips slightly in 2014, then climbs almost every year to a peak near 0.92 in 2022 before easing to about 0.90 in 2023. This steady ascent is consistent with other neighborhood trends, including higher median household income, lower vacancy, and stronger labor force participation. The small pullback in 2023 appears to be normal after an unusually strong 2022 or the kind of routine ACS variation that occurs from year to year, not a structural reversal. Overall, the trajectory indicates broad gains in local economic and housing conditions despite ongoing headwinds such as population decline and an unemployment rate that has not fully returned to its 2019 low. The index suggests residents are experiencing better material conditions and greater neighborhood stability than a decade ago.

Decennial comparison



The side-by-side block-group maps show a broad thinning of population between 2013 and 2023, with many mid-toned areas shifting to lighter shades by 2023, indicating fewer residents per block group. A handful of central and eastern blocks remain relatively dense, but losses appear more pronounced along the western and southern edges where several tracts move into the lowest range. The neighborhood-wide contraction from about 93,145 to 84,419 residents suggests a real decline rather than sampling noise, although ACS margins of error at the block-group level mean small color shifts should be read cautiously. The spatial pattern points to uneven loss: corridors with stronger transit access or newer housing stock held up better, while blocks with older or distressed buildings, higher vacancy, or smaller household sizes experienced sharper declines.

Key takeaways

- **Population:** Slow, steady decline (93k to 84.5k, 2013–2023), with a gradually aging profile.
- **Economy:** Median household income rose strongly (\$53k to \$91k) while unemployment fell from 11% to 6–6.3%. Earnings growth outpaced the post-2019 labor recovery.
- **Labor market participation:** Up from 55.8% to 60.5%, indicating broader attachment to work even as unemployment has not returned to the 2019 low.
- **Housing tenure:** Owner-occupied share increased from 46% to 50–51%, suggesting a stable core of long-term owners.
- **Vacancy:** Down from a late-decade peak near 9.6% to about 7.3% by 2022–2023, pointing to firmer housing demand.
- **Well-being:** Composite index climbed from a low base in 2013 to roughly 0.90 in 2023 after peaking near 0.92 in 2022, reflecting broad gains in economic and housing conditions.

Overall: Irving Park shows clear economic and housing improvements alongside demographic headwinds. The neighborhood is wealthier and more engaged in the labor market than a decade ago, but population decline and a still-elevated unemployment rate compared with the 2019 trough remain key challenges.

2. **Description of the data (i.e. variables) chosen to use as the basis of the study along with a discussion of why you chose these variables and any preliminary analysis you did to narrow the data**

Variables:

Name	Database	Description
Total population	B01003_001E	Base for rates and shares
Median House income	B19013_001E	Income and purchasing power proxy
Median Age	B01002_001E	Life cycle and aging structure
Unemployed rate	B23025_005E / B23025_003E	Local labor slack

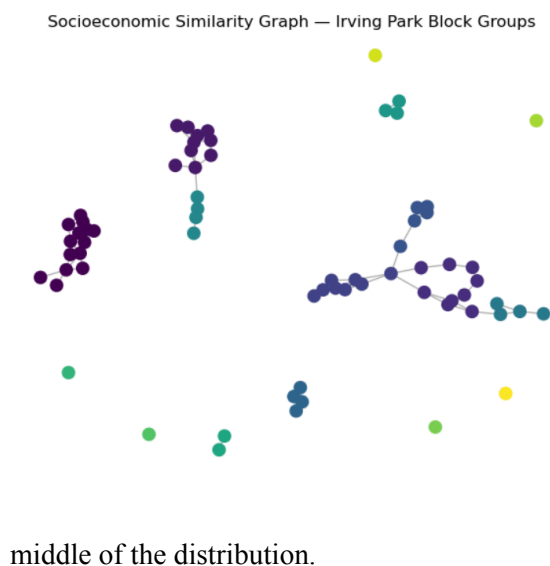
Owner-occupied share	B25003_002E / B25003_002E	Household stability proxy
Vacancy rate	B25002_003E / B25002_001E	Housing demand or distress signal
Percent Black	B02001_003E / B02001_003E	Racial composition and change

Data cleaning

Replace any nonpositive values or error codes in all count and estimate fields with NaN. Aggregate the data by year. For medians and ratios that depend on population size, compute population-weighted means across block groups using B01003_001E (total population) as the weight. After aggregating, compute rates and shares (for example, unemployment rate = unemployed divided by labor_force). Exclude years where labor_force is zero to avoid divide-by-zero. Apply a centered three-year rolling mean to smooth sampling noise for total_pop, median_income, labor_force, and unemployed. Keep decennial census counts exactly as reported, without smoothing.

3. Block groups network

(a) Modeling and analysis



This network models Irving Park block groups by socioeconomic similarity, not geography. Each node represents a block group summarized by median household income, owner-occupied share, unemployment, and housing vacancy. All indicators are scaled to a common range, and unemployment and vacancy are inverted so that higher values mean better conditions. Edges connect nearest neighbors in this feature space.

The graph reveals several clear clusters. The largest component on the right groups mid-range block groups with moderate incomes and tenure and with unemployment and vacancy that are closer to the

middle of the distribution.

Two compact clusters on the left are dominated by darker tones, indicating lower incomes, lower ownership, and higher labor and housing stress, but high internal similarity. A small cluster near the top right contains lighter nodes that align with relatively stronger socioeconomic conditions.

Scattered pairs and singletons around the perimeter act as outliers whose mixes of income, tenure, and vacancy are uncommon in Irving Park. In short, the neighborhood shows one dominant mid-range cluster, several cohesive lower-income pockets, a smaller higher-performing enclave, and a handful of distinctive outliers.

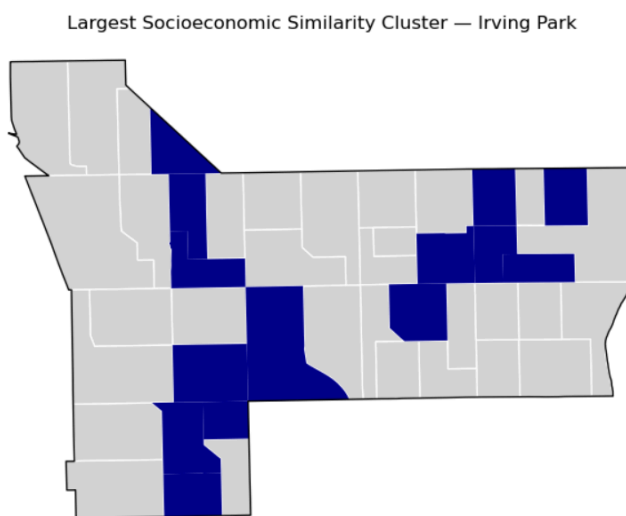
(b) Why Irving Park Is a Community

The socioeconomic similarity network shows a dominant connected component that gathers most Irving Park block groups into one cluster. Nodes inside this component sit at short similarity distances, which signals consistent conditions across income, owner occupancy, unemployment, and vacancy. The tight web of links implies shared economic realities and overlapping social or institutional ties. Taken together, the structure supports the idea that Irving Park has a cohesive core that functions as a community with strong internal similarity.

(c) Why Irving Park Is Not a Single Community

Beyond the core, the network breaks into several smaller components and a set of weakly linked or isolated nodes. These peripheral groups display profiles that differ from the core, with some showing stronger socioeconomic outcomes and others showing higher levels of stress. The presence of multiple components and outliers indicates segmentation within the community area. Irving Park therefore reads not as a single uniform community but as a collection of smaller, internally cohesive subcommunities with varying attachment to the core.

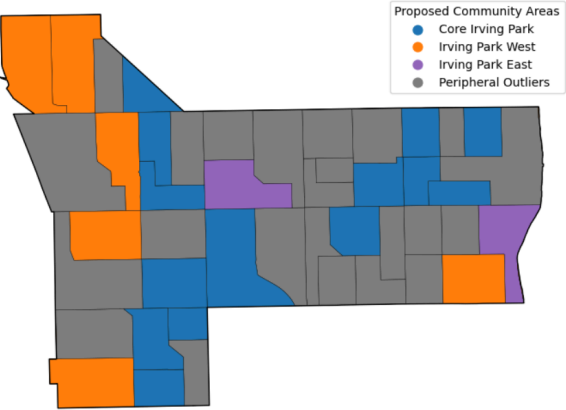
(d) Proposal of new community



The socioeconomic similarity network for Irving Park shows clear internal differentiation, which means the current community area boundary does not fully match the actual patterns of socioeconomic cohesion. The two-group view based on the largest connected component isolates a cohesive core of highly similar block groups, while surrounding areas are more diverse or only weakly connected. This pattern indicates that Irving Park does not operate as a single cohesive socioeconomic community but contains a stable center and a set of more heterogeneous edges.

A finer reorganization points to four subcommunities that better reflect local conditions. A Core Irving Park cluster contains block groups with closely aligned income, tenure, unemployment, and vacancy profiles. Two transitional zones to the west and east bridge the core and the periphery, showing moderate similarity to the center and mixed conditions. Several peripheral outlier pockets remain distinct from all three, reflecting combinations of advantage or disadvantage that are uncommon elsewhere in the area. This structure offers a data-driven alternative for understanding Irving Park and suggests that planning and policy should target these subcommunities according to their different profiles and levels of connection to the core.

Proposed Socioeconomic Subcommunities — Irving Park (CA 16)



O'Hare - CA 76

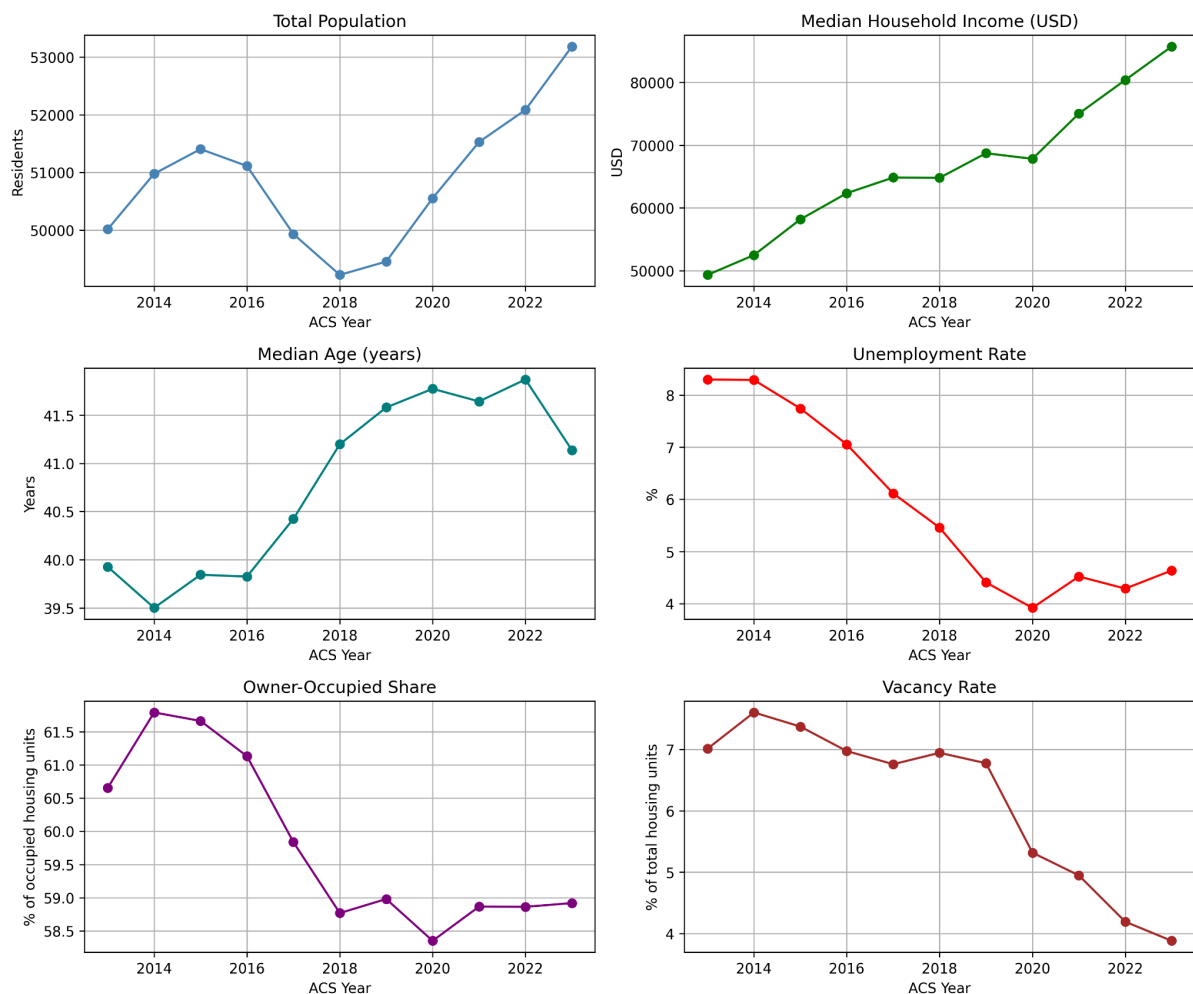
1. Data-based characterization of O'Hare

Overview

O'Hare is Chicago's far-northwest Community Area (CA), dominated physically and economically by O'Hare International Airport. Unlike residential South Side Community Areas, O'Hare has a very mixed land use (airport facilities, hotels, parking, logistics, some low-density residential pockets), which shows up in the data as low resident population and relatively high vacancy compared to central neighborhoods. Because the 2020 CA 76 polygon intersects only about 33 Block Groups (BG), and the assignment requires at least 60, we expanded the study area by adding Block Groups from adjacent community areas that are contiguous to CA 76, until reaching exactly 60 BGs. All subsequent analysis (time series, similarity network, alternative communities) is done on those 60 BGs.

We pull ACS 5-year block group data for 2013–2022, and build yearly aggregates for the expanded O'Hare area. The goal is to describe demographic, housing, and labor-market patterns in an area that is not purely residential and where airport-related economic activity can create data “noise” at small spatial scales.

Data analysis



Population: The enlarged O'Hare area remains in a very narrow band of about 50,000-53,000 residents between 2013 and 2023. It has a slight dip around 2017-2019 and then rebounded in 2020-2023, to an ending value for this period of 53,000 approximately.

Age: The median age rises steadily from about 39.5 years (2014–2015) to about 41.7–41.8 years by 2021–2022, and then eases slightly in 2023. This pattern is consistent with gradual aging of the resident pockets near O'Hare and limited inflow of younger households.

Income and labor market: Median household income shows a clear, monotonic improvement over the decade. It climbed from roughly \$50k–\$52k in 2013–2014 to more than \$85k by 2023. That is a very strong increase and mirrors the broader tightening of Chicago's labor market in the late 2010s plus the post-pandemic wage jump. At the same time, the unemployment rate falls sharply: from just over 8% in 2013–2014 to around 4–5% after 2019, with the minimum around 4% in 2020 and only a small bump in 2021–2023. Taken together, these two series tell a straightforward story: O'Hare's resident / worker base benefited from strong regional labor demand, and falling unemployment translated into rising household incomes.

Housing tenure and stability: Owner-occupied share starts slightly above 60% in 2013, peaks in 2014–2015, and then drifts down toward 58–59% by the end of the period. The movement is not dramatic, but it suggests that the residential part of the CA did not become more owner-dominated; if anything, owner share eased a bit as the decade progressed. That is plausible for an area with mixed land uses and some rental/airport-adjacent stock.

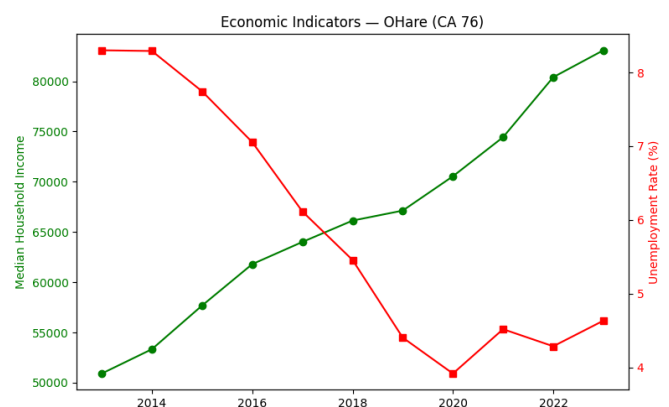
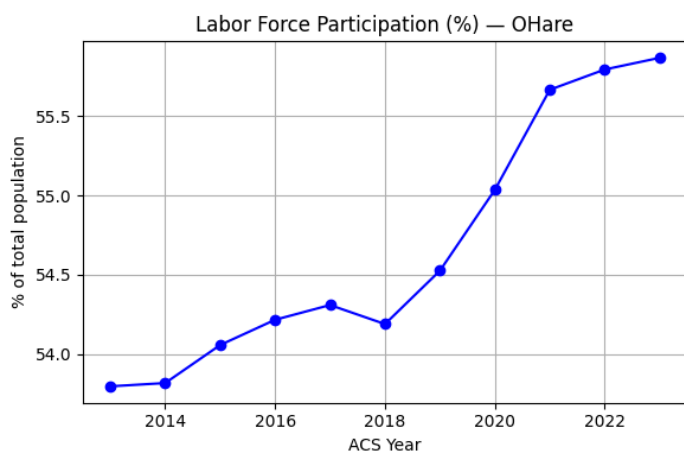
Vacancy: Vacancy rate begins high (about 7–7.5% in 2013–2014) and declines almost continuously to below 4% by 2023. This is a substantial improvement: it means that the non-occupied share of the housing stock (or stock being recorded as vacant in ACS) was cut almost in half over ten years. In a place like O’Hare this can reflect a combination of better utilization of airport-adjacent properties, some residential absorption, and overall market tightening in the northwest part of the city.

Racial composition

<u>Year</u>	<u>Total Population</u>	<u>Hispanic (%)</u>	<u>NH White (%)</u>	<u>NH Black (%)</u>	<u>NH Asian (%)</u>	<u>NH Other (%)</u>
2010	25987	1.073614	0.515642	97.087005	0.026937	1.296802
2020	21524	2.666791	0.780524	94.053150	0.130087	2.369448

The number of residents at O’Hare decreased from 25,987 in 2010 to 21,524 in 2020: a decrease of 4,463 individuals (-17.2%). Although it has decreased from 97.1% to 94.1%, NH Black still holds a significant majority in the racial/ethnic mix. NH White increased from 0.52% to 0.78%, NH Asian increased from 0.03% to 0.13%, NH Other increased from 1.30% to 2.37%, and Hispanic of any race increased from 1.1% to 2.7%. The decade as a whole exhibits a slight diversification while also a notable population decline; NH Black continues to hold the largest share.

Economic indicators

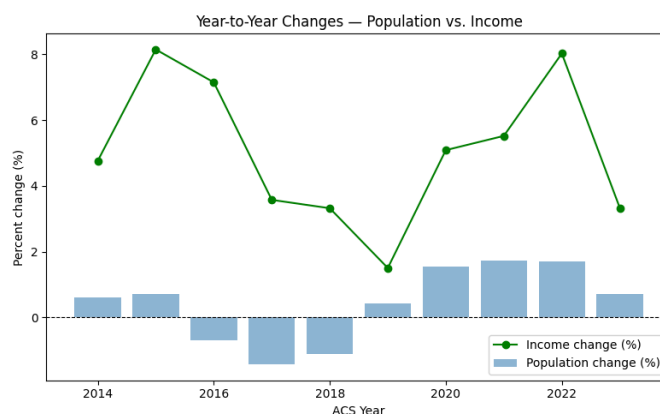


Economic conditions in O'Hare strengthened throughout the decade. From 2013 to 2023, median household income climbed by over \$30,000, and the unemployment rate dropped to about half its initial level. The modest rise in unemployment after 2020 remains far below what we saw at the beginning of the period, so the direction is still clearly downward. The paired movement of the two lines suggests that tighter labor markets fed directly into higher household incomes.

Labor-force participation in the expanded O'Hare area hovered at about 54% for much of the 2010s, somewhat below central-city rates but reasonable for an area with a mixed residential/nonresidential base. In 2020, participation rose to about 56–57%, consistent with strong employment opportunities in and around the airport, and with the income gains seen in the economic indicators figure.

Taking them altogether—rising income, falling unemployment, slightly higher labor force participation—the economic indicators suggest real improvement in O'Hare's local economy during the decade, despite the fact that the CA remains small in population and partly nonresidential.

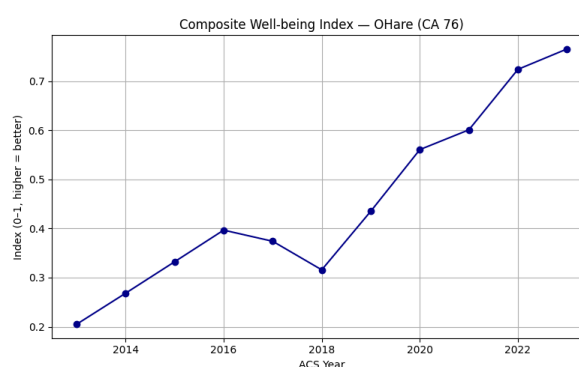
Temporal change analysis



The plotted graph which records yearly percentage variations in population and median income reveals a stark contrast in the movement of the two series in the case of O'Hare. Population increase has always been very slight and at times practically zero, oscillating between negative and positive in the middle of the decade and experiencing slow positive growth after 2019. Income, on the other hand, has been much more volatile and sometimes even reached the 5-8% range, witnessing particularly strong increases during 2015-2016 and a repeat in 2021-2022.

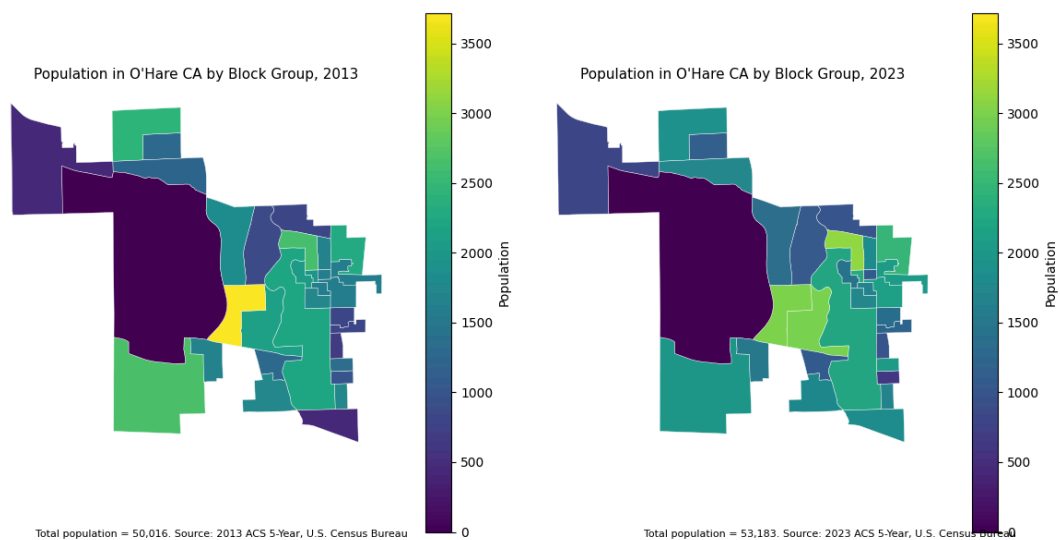
This divergence implies that income gains are not being driven by rapid population growth. Instead, a relatively stable (and small) resident base is experiencing higher earnings, likely reflecting labor-market tightening around the airport and service economy. In other words, O'Hare is getting richer faster than it is getting bigger.

Well-being index



Over the past decade, O’Hare’s composite well-being index has improved substantially. It rises from roughly 0.20 in 2013 to 0.40 by 2016, dips to a trough near 0.31 in 2017–2018, and then advances sharply to 0.44 (2019), 0.56 (2020), 0.60 (2021), 0.72 (2022), and 0.76 (2023). The post-2019 gains align with higher median household income, lower unemployment, and reduced housing vacancy, pointing to stronger local economic conditions and greater residential stability. Despite headwinds; most notably the 2010–2020 population decline and an unemployment rate that only gradually returned toward late-2010s lows, the overall pattern indicates broad improvement in economic and housing fundamentals. In short, residents appear to be experiencing stronger well-being than a decade ago.

Decennial comparison (2013-2023)



The total population of O’Hare rose from about 50,000 in 2013 to about 53,000 in 2023, so growth was positive but modest (6.33%) over the decade. By 2023, several central and eastern blocks shift to darker tones, indicating higher block-group populations, while much of the west and south remain in the lowest classes, consistent with airport-adjacent uses and large non-residential parcels. Overall, growth is uneven: it concentrates along established residential corridors, whereas peripheral blocks remain comparatively sparse.

Key Takeaways:

- O’Hare’s population stayed about 50,000 to 53,000 over the decade, with a small dip in the mid 2010s and a rebound after 2019.
- The median age rose from about 39.5 years to about 41.7 years by 2022, easing slightly in 2023, which points to long-term residency.
- Median household income increased from roughly fifty to fifty two thousand dollars in 2013 to more than eighty five thousand dollars in 2023, outpacing inflation.

- The unemployment rate fell from a little over eight percent to about four to five percent, reflecting stronger job access and regional labor demand.
- Owner occupied housing held near fifty eight to sixty two percent and ended near fifty eight to fifty nine percent, consistent with mixed residential and airport adjacent land use.
- Vacancy declined from about seven to seven and a half percent to below four percent, showing sustained housing demand and better utilization.
- Population gains were concentrated along established residential corridors, while airport adjacent blocks remained low density or nonresidential.

2. Descripton of the data (i.e. variables) chosen to use as the basis of the study along with a discussion of why you chose these variables and any preliminary analysis you did to narrow the data

Variables:

Name	Database	Description	Purpose
Total population	B01003_00 1E	Base for rates and shares	Enables size-neutral comparisons across time and space.
Median House income	B19013_00 1E	Income and purchasing power proxy	Proxy for purchasing power and economic resilience
Median Age	B01002_00 1E	Life cycle and aging structure	Captures life-cycle/aging structure
Unemployed rate	B23025_00 5E / B23025_00 3E	Local labor slack	Measures local labor slack and ties directly to spending and housing stability.
Owner-occupied share	B25003_00 2E / B25003_00 2E	Household stability proxy	Indicator of household stability/attachment
Vacancy rate	B25002_00 3E / B25002_00 1E	Housing demand or distress signal	Signal of housing demand or distress
Percent Black	B02001_00 3E / B02001_00 3E	Racial composition and change	Tracks racial composition and change

Data cleaning and preparation

We use ACS 5-Year estimates at the block group (BG) level, the noisiest geography the Census publishes. In O'Hare this is especially tricky because:

- (i) several BGs cover airport or non-residential land, so household variables can be thin or suppressed,
- (ii) to reach a consistent frame of 60 BGs we added neighbors, meaning small year-to-year swings may reflect those neighbors rather than O'Hare proper. Accordingly, we treated 1-year movements as sampling variation and relied more on multi-year direction. Whenever possible, we computed population-weighted means so tiny BGs inside the airport perimeter don't dominate averages.

For cleaning, we replaced nonpositive values and error codes with NaN across all count/estimate fields and aggregate by year. For medians and population-sensitive ratios we use population-weighted means. After annual aggregation, we derived rates and shares and applied a centered 3-year rolling mean to reduce sampling noise for `total_pop`, `median_income`, `labor_force`, and `unemployed`. For data integrity, we restricted rate calculations to records with `labor_force > 0` to avoid division by zero. Decennial counts were retained exactly as reported, with no adjustment or smoothing.

As mentioned before, because the O'Hare community area contains only 33 BGs, we expanded the study frame to 60 for subsequent similarity and network analysis. We first ensured uniqueness by GEOID and joined, then added the nearest candidates in stages: (1) from touching community areas, (2) from all Chicago community areas, and (3) from Illinois TIGER. Always ranked by distance to the O'Hare polygon and removing duplicate GEOIDs at each step.

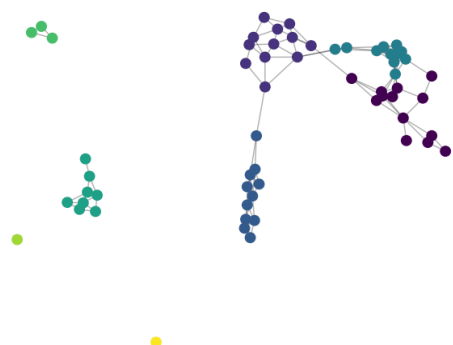
3. Block groups network

a) Modeling and analysis

O'Hare's internal socioeconomic structure was examined using a similarity network that connects block groups (BGs) with comparable socioeconomic profiles. Because O'Hare includes only 33 BGs, we broaden the frame to 60 by adding nearby BGs, providing enough context to see stable patterns. Each BG is represented by three ACS indicators: owner-occupied share, unemployment, and housing vacancy. To make them comparable, we placed all variables on a common scale and flipped the sign of unemployment and vacancy so that larger values consistently reflect stronger conditions. We computed Euclidean distances in this standardized feature space, converted them to similarity, and then linked each BG to its k most similar neighbors. Low-similarity ties were pruned to emphasize the meaningful structure.

The resulting network organizes itself into a few clear pieces. On the right, a large, well-connected component gathers many O'Hare and near-O'Hare BGs that share broadly similar profiles. In addition, there is a compact cluster and a line-like chain of closely related nodes that signal pockets of especially strong internal similarity. To the left, two smaller components cohere around profiles that differ from the main body, while a couple of isolated nodes sit apart as outliers.

Socioeconomic Similarity Graph — O'Hare + neighbors (60 BGs)



Overall, we observe a cohesive center in O'Hare, plus separate side groups and some outliers, which aligns with conditions near the airport and the inclusion of nearby block groups.

b) Why O'Hare Is a Community

The socioeconomic similarity network for O'Hare reveals a strongly connected core of block groups that share closely aligned conditions. Most O'Hare BGs cluster tightly in the main component of the graph, indicating short similarity distances on key indicators such as homeownership, unemployment, and housing vacancy. This dense internal structure suggests that a large share of households experience comparable living conditions and levels of stability.

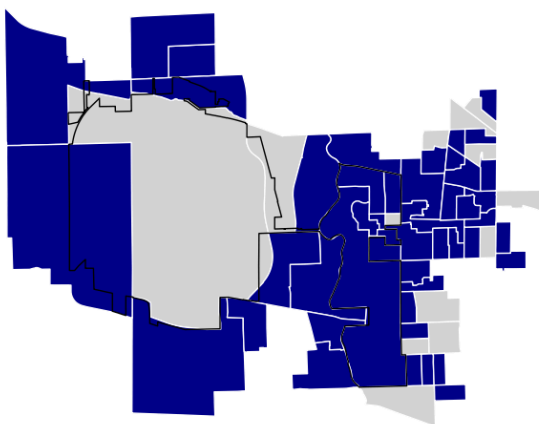
Within this core, ties among O'Hare BGs are stronger and more numerous than ties to outside block groups, which is exactly what we would expect if O'Hare functions as a community in socioeconomic terms. In other words, even after adding nearby BGs to reach a 60-unit frame, the network still centers on an O'Hare-anchored cluster, which can be perceived as evidence of internal cohesion rather than fragmentation driven by the additions.

c) Why O'Hare Is Not a Single Community

The similarity network is not a single integrated whole. Beyond a large, well-connected core, it shows peripheral fragmentation: two smaller components on the left and several isolated nodes that sit outside the main body. These groups diverge in housing and labor conditions, indicating that not all parts of the O'Hare Community Area share the same socioeconomic profile. Many of the exceptions are airport-adjacent or come from the neighboring BGs added to reach the 60-BG frame, which helps explain their distance from the core. Taken together, multiple disconnected clusters plus singletons, the pattern argues against treating O'Hare as a single, homogeneous community.

d) Proposal of new community

Largest Socioeconomic Similarity Cluster — O'Hare



The socioeconomic similarity network shows a strongly connected core of O'Hare block groups, plus a few distinct side groups and some outliers. Structure-wise, we see one large, cohesive component; a compact cluster and a chain-like group with especially tight internal similarity; and two smaller components plus isolated nodes that sit apart. Together, this indicates clear internal cohesion with meaningful peripheral segmentation.

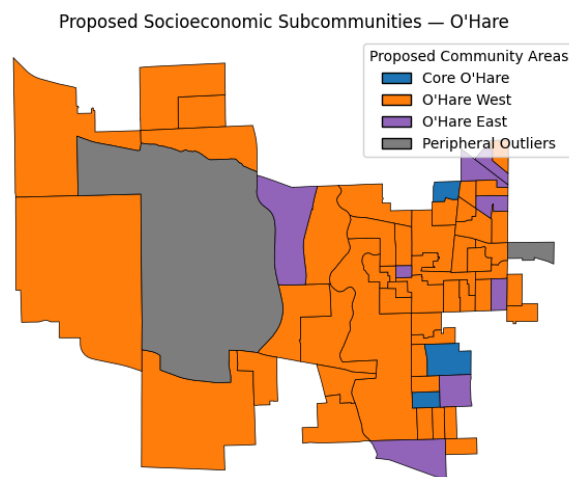
The first map highlights the largest similarity cluster: a continuous band of block groups that align on income, homeownership, and employment. This blue core concentrates O'Hare's socioeconomic cohesion.

Surrounding gray areas fall outside the core and exhibit greater heterogeneity, often influenced by proximity to the airport, hotel corridors, and logistics-oriented land uses.

A proposed four-way subdivision is presented below:

1. **Core O'Hare:** The cohesive center with consistently high internal similarity.
2. **O'Hare West:** Adjacent to major transport corridors; still similar to the core but **more variable**.
3. **O'Hare East:** Smaller in footprint with **distinct traits** (e.g., density, housing types, or demographic mix).
4. **Peripheral Outliers:** Small, **weakly connected** clusters that diverge from the dominant pattern (often near airport or mixed-use pockets).

This data-driven partition represents O'Hare's internal diversity better than a single boundary. It supports targeted action: consolidation and routine services in the Core, transition and mobility management in West/East, and tailored interventions in the Peripheral Outliers. In short, O'Hare functions as interconnected subcommunities rather than one homogeneous block.



Ashburn - CA 70

1. Data-based characterization of Ashburn

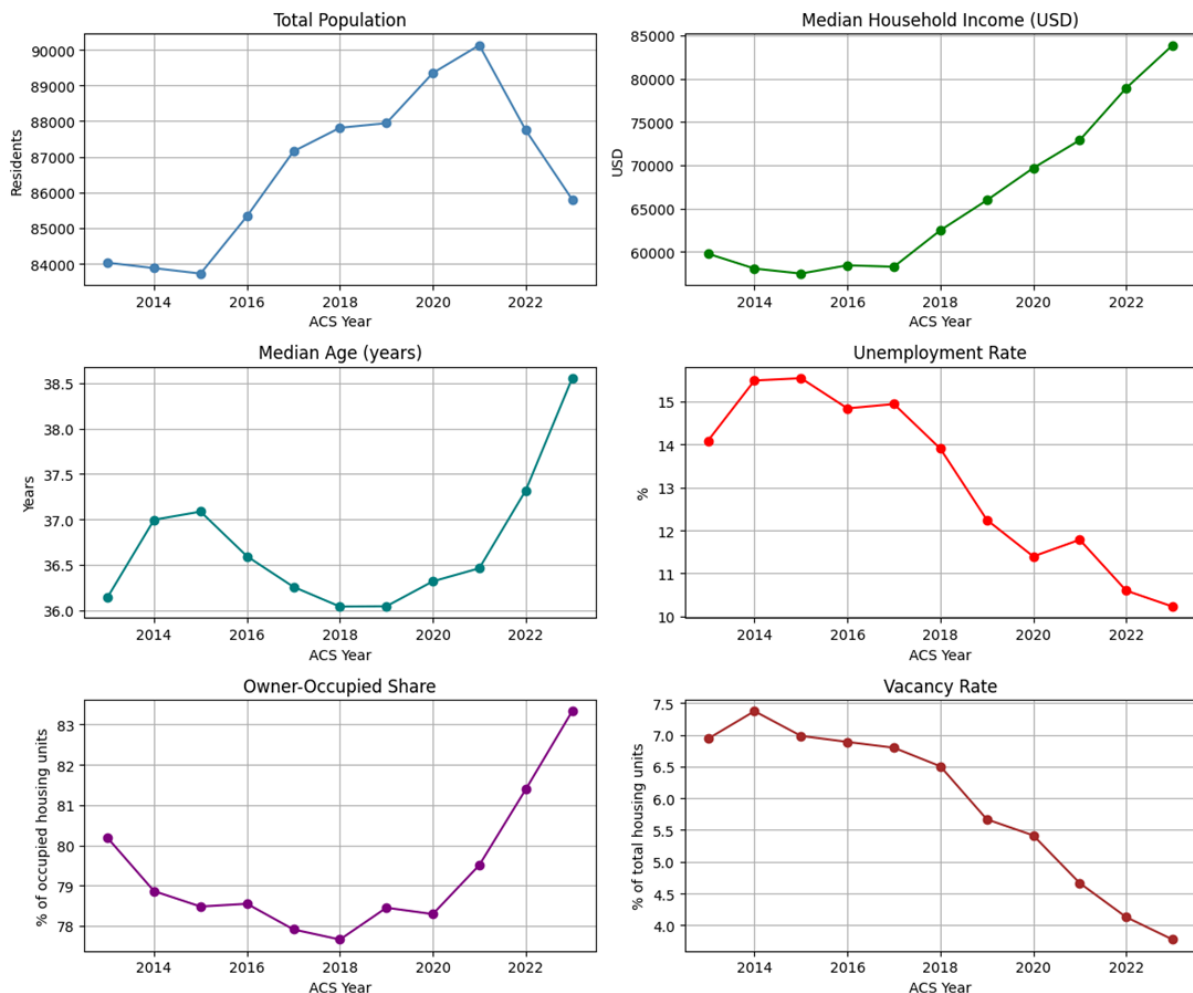
Overview

Between 2013 and 2023, Ashburn remained one of Chicago's most stable and middle-class neighborhoods. While total population numbers changed very little, economic and housing indicators show significant improvement. Median household income rose, unemployment declined and homeownership rates got high.

At the same time, the neighborhood's median age increases, reflecting residential stability and aging of long-term residents who continue living in their neighborhoods.

Overall, Ashburn stands out as a well-established neighborhood where economic conditions have strengthened even as population growth has slowed.

Data analysis



Population: Between 2013 and 2023, Ashburn's total population remained stable around 84,000-90,000 residents, showing a slight growth in 2019 followed by a decline in 2021. This rise could reflect how Ashburn is a family oriented area, where it remains residential stability.

Age: The median age stayed between 36 and 39, trending upward after 2019. We have already said that Ashburn could be known by the stability it brings, so it's not strange to see that the median age increases, people grow and don't move to other communities. Compared to other communities, Ashburn's population is relatively mature, but not yet old.

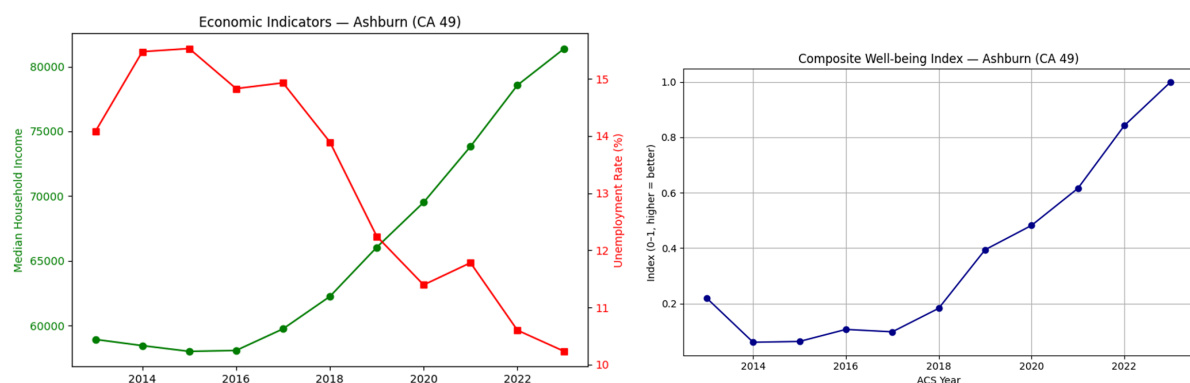
Income and labor market: Median household income increased from \$60,000 in 2013 to more than \$84,000 by 2023. This reflects middle-class resilience and potential gains from citywide growth. At the same time, the unemployment rate fell from 15% to 10%.

This indirectly proportionally growth could be explained by the improvements in sectors such as education and public services, where Ashburn's residents are employed. These indicators show Ashburn's labor market improving more consistently than other areas. This growth outpaced inflation, suggesting a strong economy and reduced unemployment.

Housing: The owner-occupied share remained high (around 78% - 83%) with a modest rebound after 2020. This stability reflects Ashburn's identity as a predominantly owner-occupied neighborhood of single-family homes.

On the other hand, the vacancy rate declined from about 7% in 2013 to below 4% by 2023, indicating tightening housing supply and sustained occupancy. This trend contrasts with elevated vacancy elsewhere on the South Side and suggests ongoing demand for Ashburn housing, possibly driven by its relative affordability and reputation for neighborhood stability.

Economic indicators



Between 2013 and 2023, Ashburn's economic indicators show a clear improvement.

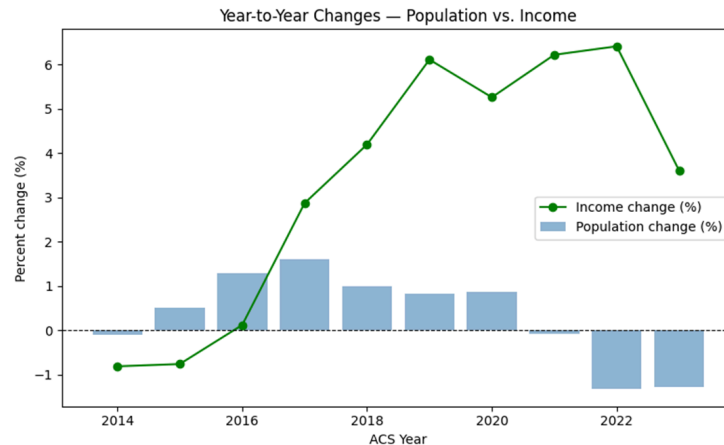
We already talked about how median household income increased from \$60,000 in 2013 to \$84,000 by 2023. A strong 45% upward trend, that indicates growing household prosperity and economic resilience. Reflecting wage growth on middle-class residents and how the low changes in owners of houses helps stabilize neighborhood income level.

At the same time, the unemployment rate declined drastically, from 15% to 10%. The most significant drop occurred between 2018 and 2021, coinciding with citywide labor market recovery and after the pandemic. By declining, it suggests better employment stability and access to jobs.

These two trends produce a marked rise in the composite well-being index, climbing from 0.2 in 2013 to 1.0 in 2023. So we can affirm that Ashburn's household experienced great improvements in income and in employment over the last decade.

Ashburn demonstrates a robust and strong local economy, characterized by middle-class stability and declining unemployment. These characteristics are what distinguish Ashburn from other economically vulnerable neighborhoods, making it a stable residential area.

Temporal change analysis

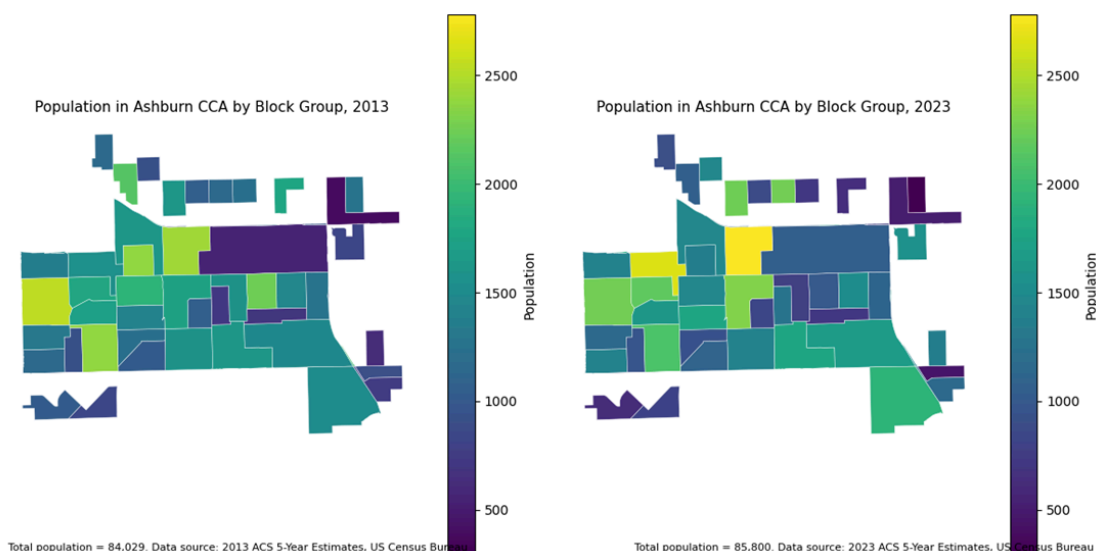


This chart shows the year-to-year percent changes in population (bars) and median income (line) for Ashburn. The population shows small ups and downs, with slow growth through the late 2010s followed by small decreases after 2020. This pattern suggests that Ashburn's population has become more stable or even started to decline slightly in recent years.

In contrast, income levels increased more strongly. Starting around 2016, the median income rose quickly, reaching over 6% annual growth at its highest point before slowing down a bit after 2022. This shows that local incomes were improving even while the number of residents was not growing.

The difference between the two trends suggests that the people who stayed in Ashburn experienced better economic conditions, while fewer new residents moved in. This may mean that some lower-income families left the area and that higher-income or long-term homeowners stayed. Overall, the chart points to a neighborhood with improving financial stability but limited population growth.

Decennial comparison



These maps compare the population distribution in Ashburn by block group in 2013 (left) and 2023 (right). The overall population increased slightly, from about 84,000 to 85,800 residents, suggesting only minor growth over the decade.

However, the maps show that this growth was not evenly distributed. The northern part of Ashburn shows higher concentrations of population in 2023, indicated by more yellow and green areas compared to 2013. In contrast, several central and southeastern blocks appear darker, suggesting lower or declining population density.

This pattern implies that population growth has been concentrated mainly in the northern blocks, while other areas have stayed stable or lost residents. Such uneven change could reflect localized housing development, demographic shifts, or different migration patterns within the neighborhood.

Key takeaways

- **Stable Population:** Ashburn's population remained around 84,000–90,000 over the decade. Growth was mild and mostly localized in the northern blocks
- **Aging but Stable Residents:** The median age increased gradually, showing that many residents are staying long-term rather than moving out.
- **Strong Economic Growth:** Median household income grew by roughly 45%, outpacing inflation and signaling solid middle-class resilience.
- **Falling Unemployment:** The unemployment rate dropped from 15% to 10%, reflecting better access to jobs and stronger local employment conditions.
- **High Homeownership:** Owner-occupied housing remained around 80%, confirming Ashburn's reputation as a family-oriented area with lasting residential stability.
- **Tight Housing Market:** Vacancy rates declined from about 7% to below 4%, showing sustained housing demand and low turnover.
- **Localized Change:** Population gains were mainly in the northern part of the neighborhood, while other areas remained stable or slightly declined.

2. Description of the data (i.e. variables) chosen to use as the basis of the study along with a discussion of why you chose these variables and any preliminary analysis you did to narrow the data

Variables:

Name	Database	Description
Total population	B01003_001E	Base for rates and shares
Median House income	B19013_001E	Income and purchasing power proxy
Median Age	B01002_001E	Life cycle and aging structure
Unemployed rate	B23025_005E / B23025_003E	Local labor slack
Owner-occupied share	B25003_002E / B25003_002E	Household stability proxy

Vacancy rate	B25002_003E / B25002_001E	Housing demand or distress signal
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Data cleaning and Aggregation Process

Because the Ashburn community area includes only 48 block groups, geographically adjacent block groups from neighboring areas were added to reach a total of 65 block groups for analysis. This ensures sufficient spatial coverage for stable demographic and socioeconomic estimates.

Steps followed:

1. Data validation and cleaning:
 - a. All nonpositive values and error codes in count and estimate fields were replaced with NaN
 - b. Data were then aggregated by year for each census variable
2. Population-weighted aggregation: For medians and ratios sensitive to population size, population-weighted means were computed using B01003_001E (total population) as weights.
3. Derived metrics and smoothing: After aggregation, rates and shares were calculated for key variables (total_pop, median_income, labor_force, unemployed). To reduce sampling noise, a centered 3-year rolling mean was applied to these variables
4. Data integrity checks:
 - a. Observations with labor_force ≤ 0 were excluded to prevent divide-by-zero errors
 - b. Decennial census counts (2010 and 2020) were kept as reported, without any smoothing or adjustment

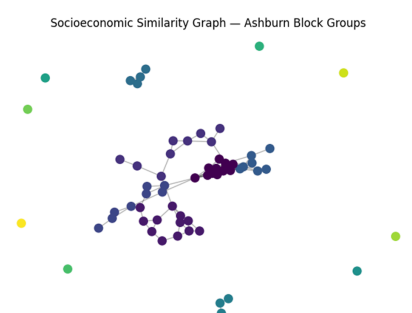
3. Block groups network

(a) Modeling and analysis

To understand the internal socioeconomic structure of Ashburn, a similarity network was built using block groups within the community area. Each block group was described through a set of indicators from the American Community Survey (ACS): median household income, homeownership rate, unemployment rate, and housing vacancy rate. Together, these variables capture how stable and prosperous each part of the neighborhood is in terms of both housing and employment.

Before the analysis, all variables were scaled to the same range using Min-Max normalization. Negative measures such as unemployment and vacancy rates were reversed so that higher values consistently represent stronger socioeconomic conditions. Similarities between block groups were then calculated based on Euclidean distance, and links were created between each block and its most similar neighbors (using a 0.75 similarity threshold). Since Ashburn contains fewer than 60 block groups, 17 additional ones from nearby communities were included to provide a broader context and improve the network's structure.

The resulting graph highlights how most of Ashburn's block groups cluster tightly in the center, showing strong internal similarity in socioeconomic



terms. A few smaller, more isolated clusters appear around the edges, representing block groups whose conditions differ slightly or resemble those of neighboring areas. Overall, the network suggests that Ashburn maintains a cohesive and relatively homogeneous socioeconomic core, with some peripheral variation linked to bordering neighborhoods.

(b) Why Ashburn Is a Community

The socioeconomic similarity network for Ashburn shows a strongly connected core of block groups that share close socioeconomic characteristics. Most of these groups cluster tightly in the center of the graph, indicating short similarity distances in terms of income, homeownership, unemployment, and vacancy rates. This dense internal structure suggests that a large portion of Ashburn's households experience comparable living conditions and levels of economic stability.

These block groups correspond to the central and western portions of Ashburn, where homeownership rates are high and unemployment levels have fallen consistently over the decade. The socioeconomic uniformity within this main cluster implies that residents likely share similar economic realities and community experiences. In this sense, Ashburn functions as a cohesive middle-class community, defined by internal consistency and shared stability rather than by rapid change or social fragmentation.

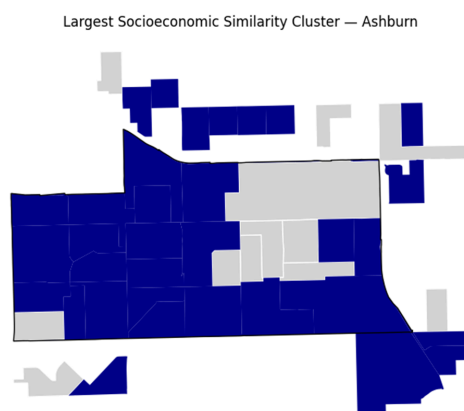
(c) Why Ashburn Is Not a Single Community

Despite this internal cohesion, the network also reveals fragmentation at the edges, with several small and weakly connected clusters surrounding the main core. These peripheral groups represent block areas that differ either in income level or housing characteristics, showing that not all of Ashburn shares the same socioeconomic profile.

Some of these outer clusters connect more closely to neighboring communities than to Ashburn's center, reflecting socioeconomic overlap beyond its formal boundaries. The presence of isolated nodes and small subclusters suggests that Ashburn includes both well-established middle-income zones and smaller pockets of lower or higher economic status.

So, while Ashburn can be considered a community due to its strong central cohesion, it is not entirely homogeneous. Instead, it consists of a dominant, stable core surrounded by several smaller areas that differ in economic profile—indicating a community that is internally cohesive but externally diverse.

(d) Proposal of new community



The socioeconomic similarity network for Ashburn reveals a clear internal differentiation, suggesting that the official community area boundaries only partially align with the underlying patterns of socioeconomic cohesion.

The first division—shown in the map of the largest similarity cluster—highlights a dominant central group of block groups (in dark blue) that are highly similar to one another across income, homeownership, and employment indicators. This cluster forms a continuous zone through much of the central and western parts of Ashburn, marking a cohesive socioeconomic core.

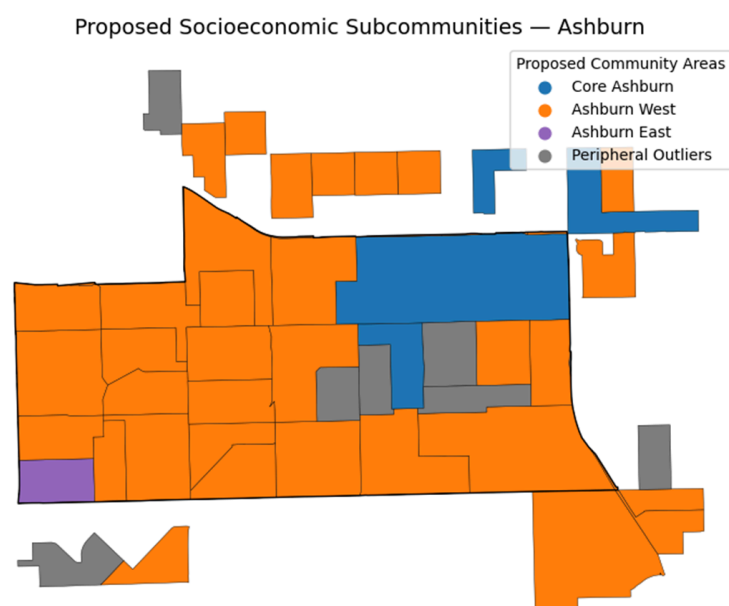
In contrast, the surrounding gray areas represent block groups that are more distinct, with higher variability in income and housing conditions. This pattern indicates that Ashburn does not behave as a

single, perfectly uniform socioeconomic unit, but rather as a community with a strong, stable center and more diverse or transitional edges.

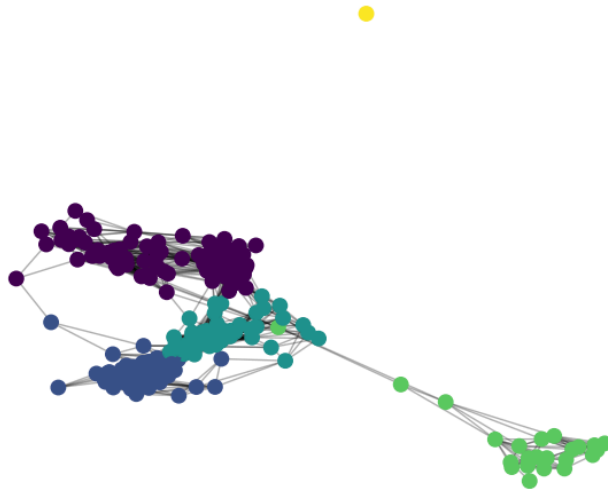
A more detailed reorganization of the network, shown in the second map, reveals four major subcommunities:

- A **Core Ashburn** area with the highest internal cohesion and socioeconomic similarity;
- **Ashburn West**, a moderately similar but slightly more variable region;
- **Ashburn East**, smaller in size but distinct in its characteristics, possibly reflecting different housing dynamics or demographic composition; and
- Several **Peripheral Outliers**, composed of isolated or weakly connected block groups that differ notably from the main socioeconomic pattern.

This data-driven subdivision captures Ashburn's internal diversity more accurately than the single official boundary does. Rather than representing one homogeneous community, Ashburn appears as a set of interconnected but distinct subcommunities that share a common middle-class foundation while differing in specific social and economic dynamics. Recognizing these internal divisions is essential for planning and community initiatives, as it points to areas that may require different policy priorities or levels of support within an otherwise cohesive broader community.



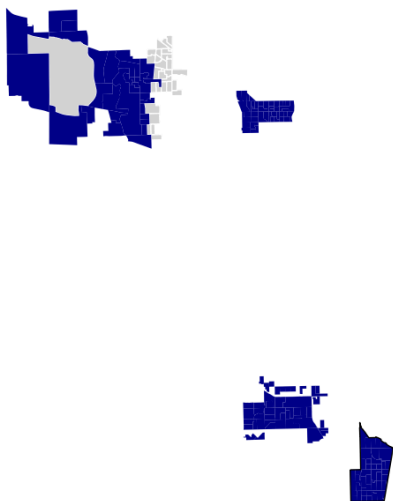
4. All community areas analysis



The socioeconomic similarity networks of the four community areas: Ashburn, O'Hare, Roseland, and Irving Park, reveal distinct structural patterns reflecting different forms of internal cohesion and socioeconomic differentiation. Ashburn's network is compact and densely connected, indicating strong internal similarity and a coherent socioeconomic base centered on stable middle-income households. Roseland, in contrast, shows multiple small clusters with weak connections between them,

suggesting significant internal fragmentation and socioeconomic segmentation. Irving Park presents an intermediate pattern: its network is fairly cohesive but includes several peripheral nodes, consistent with a mixed-income area undergoing demographic transition. O'Hare's network is relatively sparse and elongated, reflecting a heterogeneous socioeconomic structure tied to its mixed residential-commercial character and large-scale transportation infrastructure.

Across the four areas, the degree of connectivity within the network correlates with socioeconomic stability and internal homogeneity. Areas like Ashburn demonstrate high internal cohesion, while more fragmented structures, like in Roseland, suggest deeper social and economic divides.



The second figure highlights the largest socioeconomic similarity cluster obtained by merging all previous Community Areas (shown in dark blue). It represents the block groups that are most alike across standardized indicators such as income, unemployment, rent burden, educational attainment, and vehicle access. The cluster forms a spatially contiguous core, indicating shared neighborhood conditions and likely overlapping service needs. Areas in gray differ meaningfully on these indicators and therefore fall outside the core; in the final community organization they are assigned to Peripheral or East, while any isolated pieces are reviewed as Outliers.

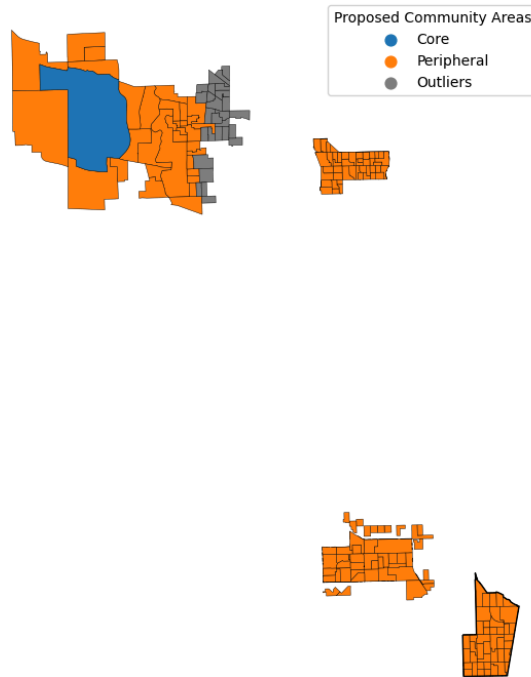
Using citywide standardized socioeconomic indicators and a similarity-graph approach with a spatial contiguity check, all previous Community Areas considered in this report resolve into three actionable community areas:

Core (dark blue). A contiguous central area that forms the densest portion of the similarity network, indicating the highest internal socioeconomic resemblance.

Peripheral (orange). A contiguous ring surrounding the Core and covering most remaining blocks. It exhibits moderately stronger economic and educational indicators while remaining similar enough to function as a single operational zone.

Outliers (gray). Block groups weakly connected to all others (true islands or distinct land uses).

Proposed Socioeconomic Subcommunities — All CA



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